📏 Depth Camera Distance Measurement (Depth Camera Distance Measurement)

1. Course Content

Using the Angstrong Nuwa-HP60C 3D depth camera, obtain the 16-bit depth image stream (`/ascamera_hp60c/camera_publisher/depth0/image_raw`), render the depth data with pseudo-color on the screen and display the center point in real-time, and show the physical straight-line distance (in millimeters mm) of a mouse-clicked point in 3D space.

> **Important**
>
> This course is only applicable to users with the **Angstrong Nuwa-HP60C Depth Camera** version.

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2. Preparation and Running

This case requires opening 2 terminals to start the camera driver and distance measurement node respectively:

2.1 Start Camera Low-level Driver

Run in **Terminal 1**:

```
ros2 launch dofbot_pro_driver A1pro_camera_arm.launch.py
```

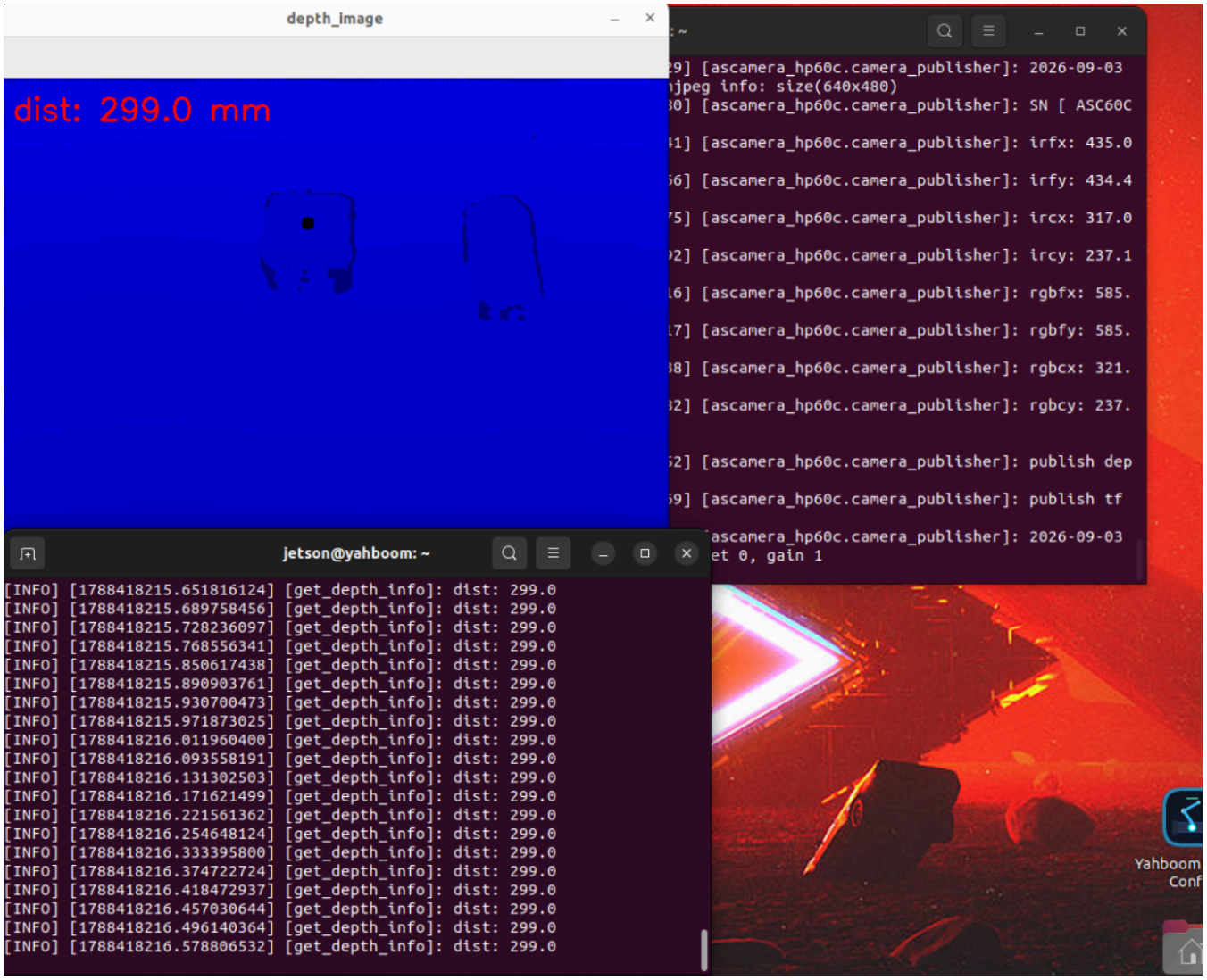
```
raSrv.cpp] [112] [onAttached] attached
[camera_launch-1] [ascamera_node-1] [INFO] [1788418150.121532507] [ascamera_hp60c.camera_publisher]: 2026-09-03 14:49:10[INFO] [Came
raSrv.cpp] [149] [onAttached] this is a new attach device, create and open it
[camera_launch-1] [ascamera_node-1] [INFO] [1788418151.131593361] [ascamera_hp60c.camera_publisher]: 2026-09-03 14:49:11[INFO] [Came
raSrv.cpp] [158] [onAttached] get model type 9
[camera_launch-1] [ascamera_node-1] [INFO] [1788418151.133700898] [ascamera_hp60c.camera_publisher]: 2026-09-03 14:49:11[INFO] [Came
raSrv.cpp] [352] [getConfigFile] get file: /home/jetson/yahboomcar_ros2_ws/software/library_ws/src/ascamera/configurationfiles/hp60c
_v2_01_20241104_configEncrypt.json
[camera_launch-1] [ascamera_node-1] [INFO] [1788418151.133843461] [ascamera_hp60c.camera_publisher]: 2026-09-03 14:49:11[INFO] [as_c
amera_sdk_api.cpp] [276] [AS_SDK_GetCameraAttrs] pid:26403  vid13442 busnum: 1  dev:4  port:1  ports:2.1
[camera_launch-1] [ascamera_node-1] [INFO] [1788418151.133885702] [ascamera_hp60c.camera_publisher]: usb camera
[camera_launch-1] [ascamera_node-1] [INFO] [1788418151.133912902] [ascamera_hp60c.camera_publisher]: bnum:1
[camera_launch-1] [ascamera_node-1] [INFO] [1788418151.133934759] [ascamera_hp60c.camera_publisher]: dnum:4
[camera_launch-1] [ascamera_node-1] [INFO] [1788418151.133963463] [ascamera_hp60c.camera_publisher]: port_numbers:2.1
[camera_launch-1] [ascamera_node-1] [INFO] [1788418151.133986152] [ascamera_hp60c.camera_publisher]: create a new publisher info set
[camera_launch-1] [ascamera_node-1] [INFO] [1788418151.144105519] [ascamera_hp60c.camera_publisher]: 2026-09-03 14:49:11[INFO] [as_c
amera_sdk_api.cpp] [276] [AS_SDK_GetCameraAttrs] pid:26403  vid13442 busnum: 1  dev:4  port:1  ports:2.1
[camera_launch-1] [ascamera_node-1] [INFO] [1788418151.145374475] [ascamera_hp60c.camera_publisher]: 2026-09-03 14:49:11[INFO] [TfTr
eeFrameIdInfo.cpp] [11] [TfTreeFrameIdInfo] Namespace /ascamera_hp60c
[camera_launch-1] [ascamera_node-1] [INFO] [1788418151.236050269] [ascamera_hp60c.camera_publisher]: 2026-09-03 14:49:11[INFO] [Came
raHp60c.cpp] [1830] [getConfigurationParameters] according to the configuration file.
[camera_launch-1] [ascamera_node-1] [INFO] [1788418151.236447014] [ascamera_hp60c.camera_publisher]: 2026-09-03 14:49:11[INFO] [Came
raHp60c.cpp] [1535] [parseConfigFileParameter] moduleName:AS_CAM_HP60C
[camera_launch-1] [ascamera_node-1] [INFO] [1788418151.236634858] [ascamera_hp60c.camera_publisher]: 2026-09-03 14:49:11[INFO] [Came
raHp60c.cpp] [1562] [parseConfigFileParameter] Parsing config file /home/jetson/yahboomcar_ros2_ws/software/library_ws/src/ascamera/
configurationfiles/hp60c_v2_01_20241104_configEncrypt.json
[camera_launch-1] [ascamera_node-1] [INFO] [1788418151.236952626] [ascamera_hp60c.camera_publisher]: 2026-09-03 14:49:11[INFO] [Came
raHp60c.cpp] [1627] [parseConfigFileParameter] configuration protocol Version: v2.x
[camera_launch-1] [ascamera_node-1] [INFO] [1788418151.236997075] [ascamera_hp60c.camera_publisher]: 2026-09-03 14:49:11[INFO] [Came
raHp60c.cpp] [1638] [parseConfigFileParameter] configuration Version: v2.0.1.20241104
[camera_launch-1] [ascamera_node-1] [INFO] [1788418151.281295908] [ascamera_hp60c.camera_publisher]: 2026-09-03 14:49:11[INFO] [Came
raHp60c.cpp] [1803] [setParamersAfterOpenCam] Camera confiParaEnable true, setting configuration parameter
[camera_launch-1] [ascamera_node-1] [INFO] [1788418152.770157833] [ascamera_hp60c.camera_publisher]: 2026-09-03 14:49:12[ERROR] [XuC
mdCameraHp60c.cpp] [454] [getMjpegSize] get mjpeg from xu command failed
[camera_launch-1] [ascamera_node-1] [INFO] [1788418152.770322861] [ascamera_hp60c.camera_publisher]: 2026-09-03 14:49:12[INFO] [XuCm
dCameraHp60c.cpp] [458] [getMjpegSize] mjpeg size:640x480
[camera_launch-1] [ascamera_node-1] [INFO] [1788418152.770439344] [ascamera_hp60c.camera_publisher]: camera opened
[camera_launch-1] [ascamera_node-1] [INFO] [1788418152.770478161] [ascamera_hp60c.camera_publisher]: get config info, ret 0, is_Regi
stration 1
[camera_launch-1] [ascamera_node-1] [INFO] [1788418152.771961299] [ascamera_hp60c.camera_publisher]: #camera[0xaaaaba795940] SN[ASC6
0CF19001984]'s firmware version:DLS_VERc20865=:V01.09_2024110714:V01.19_2024110714:L01
[camera_launch-1] [ascamera_node-1] [INFO] [1788418152.773051531] [ascamera_hp60c.camera_publisher]: set depth resolution: 640 x 480
 @ 25fps
[camera_launch-1] [ascamera_node-1] [INFO] [1788418152.774024866] [ascamera_hp60c.camera_publisher]: set rgb resolution: 640 x 480 @
 25fps
[camera_launch-1] [ascamera_node-1] [INFO] [1788418152.783542587] [ascamera_hp60c.camera_publisher]: 2026-09-03 14:49:12[INFO] [Came
raHp60c.cpp] [259] [startStreaming] start streaming
[camera_launch-1] [ascamera_node-1] [INFO] [1788418152.783690974] [ascamera_hp60c.camera_publisher]: 2026-09-03 14:49:12[INFO] [Came
raSrv.cpp] [206] [onAttached] attached end
[camera_launch-1] [ascamera_node-1] [INFO] [1788418153.786566161] [ascamera_hp60c.camera_publisher]: 2026-09-03 14:49:13[INFO] [Came
raHp60c.cpp] [278] [stopStreaming] stop streaming
```

2.2 Start Depth Distance Measurement Node

Run the distance measurement application node in **Terminal 2**:

```
ros2 run yahboomcar_depth get_center_dis
```

Click on the image to select a point to get the corresponding distance. You can control the robotic arm yourself if you want other arm angles.



3. Operation Flow and Interaction Description

1. After the node starts, it will automatically pop up the `depth_image` pseudo-color visualization window;
2. By default, measure the depth distance of the center point (x=320, y=240) of the screen, and display `dist: xxx.x mm` in real-time in the upper left corner of the image;
3. **Mouse Click Distance Measurement**: Left-click any object area in the pop-up window, the black circle marker will automatically move to the clicked position, and the distance value in the upper left corner will immediately update to the real physical distance corresponding to that target pixel;
4. The terminal window will synchronously print the current click coordinates and depth measurement logs

> **Note**
>
> The effective measurement range of the depth camera is $200\text{mm} \sim 2500\text{mm}$ (0.2m~2.5m). If the target distance is less than 200mm (entering the measurement blind zone) or exceeds 2.5m, the measured value may be 0 or NaN.

4. Core Source Code Analysis

Source Code Path:

```
$HOME/yahboomcar_ros2_ws/yahboomcar_ws/src/yahboomcar_depth/yahboomcar_depth/Basic/get_center_dis.py
```

```
class GetDepthInfo(Node):
    def __init__(self):
        super().__init__('get_depth_info')
        # 1. Subscribe to Nuwa depth image topic (16UC1 millimeter unit encoding)
        self.sub = self.create_subscription(
            Image, '/ascamera_hp60c/camera_publisher/depth0/image_raw', self.topic_callback,
10)
        self.window_name = 'depth_image'
        self.y = 240
        self.x = 320
        self.depth_bridge = CvBridge()

    def click_callback(self, event, x, y, flags, params):
        # 2. Mouse left button click event capture
        if event == 1:
            self.x = x
            self.y = y
            self.get_logger().info(f'Clicked coordinates: x={self.x}, y={self.y}')

    def topic_callback(self, msg):
        # 3. Convert ROS Image message to 32-bit floating point matrix
        depth_image = self.depth_bridge.imgmsg_to_cv2(msg, '32FC1')
        frame = cv.resize(depth_image, (640, 480))
        depth_image_info = frame.astype(np.float32)

        # 4. Read the physical depth value at the specified pixel coordinates (unit:
millimeter)
        dist = depth_image_info[self.y, self.x]

        # 5. Pseudo-color rendering (JET color mapping, red for near, blue for far)
        depth_image_colored = cv.applyColorMap(
            cv.convertScaleAbs(depth_image, alpha=0.06), cv.COLORMAP_JET)

        # 6. Draw measurement info and sample point marker
        dist_text = f'dist: {round(dist, 3)} mm'
        cv.putText(depth_image_colored, dist_text, (10, 40), cv.FONT_HERSHEY_SIMPLEX, 1.0,
(0, 0, 255), 2)
        cv.circle(depth_image_colored, (self.x, self.y), 1, (0, 0, 0), 10)
        cv.imshow(self.window_name, depth_image_colored)
        cv.setMouseCallback(self.window_name, self.click_callback)
        cv.waitKey(1)
```